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Automotive:

“AI can increase efficiency... to about 90 per cent”

The rise of autonomous vehicles calls for improving the systems regularly. Every kilometre that an autonomous vehicle covers, delivers valuable data in its development. Puneet Gupta, chief technology officer, Brillio, talks about the challenges involved in autonomous vehicle development with Saurabh Durgapal of EFY

Q. What are the trends in automotive testing?

A. In the early stages of system development while building a core system, it is important to invest in building emulators. This helps in managing testing in the early stages and reduces costs. Cost of corrections increases significantly if you go for testing after the system has been developed. Use of artificial intelligence (AI) in creating user modelling and auto creation of new test scenarios is a high-potential area.

Q. How does AI help with testing?

A. While testing any autonomous system manually, the number of test cases are limited by the imagination of the tester in charge; whereas, a car can be driven by someone with a much different notion of driving. AI, on the other hand, can model various kinds of drivers based on their driving habits, and can give a better test case scenario. Adding AI can increase efficiency from 65 per cent that comes with manual testing to about 90 per cent.

Q. Please walk us through the systems in an autonomous vehicle?

A. There are three major parts in the system involved in autonomous vehicles. First is the intra-vehicle sensory network and associated edge processing, which is involved in instantaneous decision making.

Second is Cloud based analytics and control hub that process the enormous data being generated and

draw insights and behaviour-modification decisions.

Third is the external sensor and communication infrastructure that can help enhance autonomous performance and experiences.

Q. Can we rely on middleware?

A. With the right architecture, we can rely heavily on middleware. Everything can be broken down to critical, medium- and low-critical decisions. Critical decisions happen inside the on-board computing device in the vehicle. Decisions that can wait, for example, driving behaviour and optimal speed, can be sent to the Cloud. There is also the possibility of some onboard component failing. Algorithms in this case are designed to take actions depending on the car health.

Q. How do these systems learn to be safe with human life?

A. At the core is learning and risk-calculation algorithms. Initially, the systems are overly cautious, which reduces over time in their trust to imitate humans. Alternatively, they can operate in assisted mode. Once the systems learn to take calculated risks, these can graduate to autonomous mode. Algorithms are designed to work by carefully weaving in imperfections that result in more intelligent behaviours.

Q. How can modifying algorithms impact vehicles?

A. Accidents happen when we try

to stretch the boundaries of algorithms in autonomous systems, and while trying to make these behave more like human beings. As humans, while driving, we do not stop at every turn. Instead, we take a calculated risk and go ahead. Computing systems are typically built around hard choices and, hence, are not tuned to calculated risks. This is where we begin pushing the boundaries of autonomous systems to take some calculated risks.

Q. What are the future expectations in delivering end experiences?

A. A smartphone on its own can bring in smart driving experiences today. A range of sensors are already available and more are getting added every day. So if a vehicle is involved in an accident, the smartphone instead of acting dead can react on sudden deceleration data and guide the driver.

Once data is accumulated, focus is going to be on innovative uses of that data.

Q. What are the available analytics tools for testing?

A. There are some good open source and commercial options available today, but most of these are in patches and must be clubbed together for extensive usage. So anybody doing extensive usage must build some core components of their own. That is the approach we are taking as well—leverage the good open source available but invest heavily to build a core integrated foundation. **EFY**